

# PORTFOLIO

## Assignments for the course: Project Java and Web Development (DLBCSPJWD01)

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## 1. TOPICS AND TASKS

Within the framework of this course, one of the following topics must be selected.

### 1.1. Task: Develop a web application

In this project you will develop a web application of your choice. The most important outcome is that you deliver a functioning application; in other words, UI mockups and design flows are not sufficient as the main deliverable. The purpose of this project is to realize an idea in code, not on paper. Even though the overall details of the project are up to your creativity and interests, your project must meet the following criteria:

- Your application front-end should be written in HTML, CSS and JavaScript (or equivalents).
- Alternatively, you have a choice to use Web Component Frameworks, such as Java Server Faces or JavaScript based frameworks such as React.js or Vue.js as well for your front-end.
- You need to also develop a back-end in a technology of your choice (e.g. NodeJS, Java Server Faces, PHP, etc.). You can choose based on your expertise and preferences. External APIs (like Weather APIs or News APIs) can be used but should only be consumed by the backend and then transformed for the front-end. They should not be directly used from front-end. Furthermore, make sure that you provide credentials for the API (in the submission document), if it is not publicly available. We will not register for any third-party service.
- Your front-end should have at least two dynamic aspects, which means interactions with your developed backend that are developed in JavaScript.
- The front-end should be responsive, i.e. when you resize the browser to different view port dimensions or use your web application on a smartphone/tablet, the webpage must continue to be readable.
- Your front-end must communicate with the server back-end, where the back-end receives request to either send data back to the front-end or stores data in the back-end.
- The application must run in a web browser.
- The software must be appropriately documented, i.e., code documentation. Appropriate means sufficient documentation in the code for another developer to understand intent and execution.
- The design choices (technologies) should be documented.
- The source code and any necessary installation files should be exported to a folder and MUST include sufficient information for installation (provided in the README file). This folder will be zipped later and uploaded in phase 3.
- A short video / screen cast must be prepared to demonstrate the final application. The video/screencast should not exceed 2min and should show the desktop view as well as the responsive view of your application.

Implement a web application in the following 3 phases:

#### 1.1.1. Conception phase

This phase represents the most important part of the design process. In this phase you will *make design decisions* on what your web application is going to be about and how you want to build it, i.e., the technology you want to use both on the front-end and back-end of your application.

The first step is to create a *brief concept*. The concept should illustrate the purpose of the web application, target user groups, the benefit of a user using the application and the technical components/technologies required to meet the overall user requirements. It is important for you to document this *in words*. Remember, you have very

little constraints in your choices, hence you have an opportunity to play to your strength and interests with the choices you make.

Keep in mind, this is conceptual. Do not start building in this phase but think through what it is that you want to build and have a somewhat clear idea, how you want to do it.

**Your deliverable for Phase 1** is the conceptual design, that should be described in a presentation with maximum 5 slides (excluding title slide).

**For your submission**, please submit on PebblePad as a PPTX file.

Throughout the process, online tutorials are offered and they provide an opportunity to talk, share ideas and/or drafts, and obtain feedback. In the online tutorials, exemplary work can be discussed with the tutor. Here, everyone has the opportunity to get involved and learn from each other's feedback. **It is recommended to make use of these channels to avoid errors and to make improvements.** You should only submit work after making use of the above-mentioned tutorial and informative media. This will be followed by feedback from the tutor; then, the work on the second phase can begin.

### 1.1.2. Development phase/reflection phase

In this phase, you will *start implementing*. In professional software development, there are many steps to converge towards your desired end state. In the context of this project, you should focus on coding until your web application works. This will require several iterations until you achieve a state that represents the end-state you had in mind and documented in the conception phase. Specifically, in this phase, you will do the following:

- code
- fix bugs
- refine your idea from the conception phase
- Google for solutions for challenging parts of your solution
- test

In addition, you will

- upload your code in a GitHub repository. If you do not have a public GitHub account, please create one and learn how to manage your code in GitHub. (Your final submission will require that you have all your code assets organized in GitHub, hence, it is advisable to start familiarizing yourself in this phase and use GitHub to manage different iterations of your code).
- comment your code.
- document changes you may have made compared to your conception phase proposal.
- define test data and test cases for your solution (preferably coded as unit or integration tests).

**Your deliverable for Phase 2** is an extension of the phase 1 submission presentation. You should not add more than 10 slides to that presentation with the following content:

- Modify your title-slide to include the link to your public GitHub repository.
- Include an overview of final technology choices for front and backend (e.g. HTML, CSS, JavaScript + NodeJs + Express + MongoDB) – it is ok if this is the same as in phase 1 if your choices have not changed
- Include screenshots of your web application with explanations, showing the responsive view of your application as well.

- Include one slide on what you have decided to change compared to your proposal and why. If you did not change anything, please explicitly say so.
- Prepare a 1-2min screencast video of your current state of the web. Your screencast should contain a voice-over or subtitles and should also show the responsive view of your web application.
- Test cases (if not coded in software).

**For your submission**, please submit the presentation into PebblePad. Please add the link to your public GitHub repository also in the text-field in PebblePad.

Throughout the process, online tutorials and other channels provide the opportunity to profoundly discuss ideas and/or drafts and to get sufficient feedback, tips, and hints. **It is recommended to use these channels to avoid errors and to improve your work.** Once this is done, you can hand in your second phase for evaluation. Following feedback from the tutor, your work on the final draft will continue in the third phase.

### 1.1.3. Finalization phase

In the finalization phase, the goal is to *optimize the solution* after having received feedback from the tutor, and to complete the task. Certain elements may have to be improved or changed again to finalize the task and complete the portfolio course. It is important that you consider the feedback that you received on your submission at the end of the previous phase. Depending on the changes you make in your software, it might be necessary to provide an updated version of the documentation. Make sure your software comes with installation and run instructions so that it is clear how to install the application.

All essential content of your work is in your GitHub repository which contains the following deliverables:

1. Your code folders, including all code + libraries necessary to deploy and run the web app
2. Installation and run instructions in the README.md file

**Your deliverable for Phase 3** is an extended version of the already created PowerPoint presentation from phase 1 and 2. On maximum 10 additional slides, please describe your solution in terms of content and concept. This abstract presents a short break-down (“making of” of the project) about the technical approach and your development steps in a clear and informative way (i.e., how you used the technology to develop your web application). Please also include lessons learned.

#### **For your submission:**

1. Upload the updated PPTX file that covers all three phases in PebblePad.
2. Put the link to your GitHub Repository into the PebblePad text field.
3. Export your GitHub repository and put it into a folder. You must zip and upload this folder in your submission in PebblePad.

In the “Finalization phase”, the online tutorials and other channels also provide the opportunity to obtain sufficient feedback, tips, and hints before the finished product is finally handed in. **It is recommended to use these channels to avoid errors and to make improvements.** Following the submission of the third portfolio page, the tutor submits the final feedback which includes evaluation and scoring within six weeks.

## 2. TUTORIAL SUPPORT

In principle, several channels are open to attain feedback for the portfolios. The respective use is the sole responsibility of the user. The independent development of a product and the work on the respective portfolio parts is part of the examination performance and is included in the overall assessment.

On the one hand, the tutorial support provides feedback loops on the portfolio parts to be submitted in the context of the conception phase as well as the development and reflection phase. The feedback takes place within the framework of a submission of the respective part of the portfolio. In addition, regular online tutorials are offered. These provide you with an opportunity to ask any questions regarding the processing of the portfolio and to discuss other issues with the tutor. The tutor is also available for technical consultations as well as for formal and general questions regarding the procedure for portfolio management.

Technical questions regarding the use of “PebblePad” should be directed to the exam office via mail.

## 3. EVALUATION

The following criteria are used to evaluate the portfolio with the percentage indicated in each case:

| Evaluation criteria         | Explanation                                                                                                | Weighting |
|-----------------------------|------------------------------------------------------------------------------------------------------------|-----------|
| Problem Solving Techniques  | *Capturing the problem<br>*Clear problem definition/objective<br>*Understandable concept                   | 10%       |
| Methodology/Ideas/Procedure | *Appropriate transfer of theories/models<br>*Clear information about the chosen Methodology/Idea/Procedure | 20%       |
| Quality of implementation   | *Quality of implementation and documentation                                                               | 40%       |
| Creativity/Correctness      | *Creativity of the solution approach<br>*Solution implemented fulfils intended objective                   | 20%       |
| Formal requirements         | * Compliance with formal requirements                                                                      | 10%       |

The design and construction of the portfolio should take into account the above evaluation criteria, including the following explanations:

**Problem Solving Techniques:** Clear scope as well as comprehensibility of the project. Use of appropriate technologies, libraries and services, understandable technological choices and functionality of the web application.

**Methodology/Idea/Procedure:** Development decisions, recognizable design and technology decisions, meaningful choices.

**Quality of implementation:** Visible concepts of software development, clean code, adherence to consistent documentation of code, compliance with the acceptance criteria outlined in chapter 1.

**Creativity/Correctness:** Originality of solution, correctness of the solution as verified by testing.

**Formal requirements:** The submission follows the acceptance criteria from Chapter 1 and the formal guidelines following in the next chapter. It is particularly important to respect the formal submission requirements outlined in Chapter 4.

## 4. FORMAL GUIDELINES AND SPECIFICATIONS FOR SUBMISSION

### 4.1. Components of the examination performance

The following is an overview of the examination performance portfolio with its individual phases, individual performances to be submitted, and feedback stages at one glance. A template in “PebblePad” is provided for the development of the portfolio parts within the scope of the examination performance. The presentation is part of this examination.

| Stage                                                                    | Intermediate result | Performance to be submitted                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------------------------------------------------------------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Conception phase                                                         | Portfolio part 1    | <ul style="list-style-type: none"> <li>5 slides presentation (without title slide) including conceptual design in written form with: <ul style="list-style-type: none"> <li>Brief description of the envision web application, intended target user group and benefits to the user as well as technological choices</li> </ul> </li> <li>Other relevant information into PebblePad text field (limited to ½ Din A4 page) as you see fit</li> </ul>                                                                                                            |
| Feedback                                                                 |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Development phase/<br>reflection phase                                   | Portfolio part 2    | <ul style="list-style-type: none"> <li>A presentation of not more than 10 slides (in addition to the title-slide and the 5 slides from phase 1) containing: <ul style="list-style-type: none"> <li>Overview of final technology choices</li> <li>Annotated screenshots of your web application</li> <li>Changes compared to Phase 1</li> <li>1-2min screencast showing the current state in desktop and responsive mode.</li> <li>Test cases (if not coded in software)</li> </ul> </li> <li>Link to GitHub repository in the PebblePad text field</li> </ul> |
| Add GitHub repository link to title-slide                                |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Feedback                                                                 |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Finalization phase                                                       | Portfolio part 3    | <ul style="list-style-type: none"> <li>Full presentation with all slides from phase 1 and 2 as well as additional 10 slides for the abstract (“making of” and lessons learned)</li> <li>Upload the zip-folder (including the export of GitHub repository)</li> </ul>                                                                                                                                                                                                                                                                                          |
| Put the link to your GitHub Repository into the text field in PebblePad. |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Feedback + Grade                                                         |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

## 4.2. Format for Digital File Submission

### Conception phase

|                                           |                                                                |
|-------------------------------------------|----------------------------------------------------------------|
| Recommended tools/software for processing | Powerpoint                                                     |
| Permitted file formats                    | PPTX                                                           |
| File size                                 | as small as possible                                           |
| Further formalities and parameters        | Files must always be named according to the following pattern: |

**For the performance-relevant submissions on “PebblePad”:**

Name-FirstName\_MatrNo\_Course\_P(hase)-1\_S(ubmission)

Example: Mustermann-Max\_12345678\_DLBCSPJWD01\_P1\_S

### Development/reflection phase

|                                           |                                                                                                                    |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Recommended tools/software for processing | Powerpoint<br>Screencast tool (e.g. screencast-o-matic)<br>Coding platform of choice (e.g. Visual Code or similar) |
| Permitted file formats                    | PPTX (must include screencast video)                                                                               |
| File size                                 | 100MB                                                                                                              |
| Further formalities and parameters        | Files must always be named according to the following pattern:                                                     |

**For the performance-relevant submissions on “PebblePad”:**

Name-FirstName\_MatrNo\_Course\_P(hase)-2\_S(ubmission)

Example: Mustermann-Max\_12345678\_DLBCSPJWD01\_P2\_S

Please make sure that you embed the screencast video inside the PPTX as you can only upload 1 file!

### Finalization phase

|                                           |                                                                                                                                                                                                          |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Recommended tools/software for processing | Powerpoint<br>Screencast tool (e.g. screencast-o-matic)<br>Coding platform of choice (e.g. Visual Code or similar)                                                                                       |
| Permitted file formats                    | PPTX                                                                                                                                                                                                     |
| File size                                 | as small as possible                                                                                                                                                                                     |
| Further formalities and parameters        | <b>IMPORTANT</b> is the upload of the zip folder that has been created especially for the submission (please follow the instructions on myCampus). This folder contains your exported GitHub repository. |

#### The folder structure then looks like this:

- Main directory (name of the zip folder) -> *Name: Name-First\_Name\_Matriculation\_Course*, containing the exported GitHub repository  
*Example: Mustermann-Max\_DLBCSPJWD01\_PJWD*

Files must always be named according to the following pattern:

#### For the performance-relevant submissions on “PebblePad”:

Name-FirstName\_MatrNo\_ Course \_P(hase)-3\_S(ubmission)  
Example: Mustermann-Max\_12345678\_DLBCSPJWD01\_P3\_S

If you have any questions regarding the submission of the portfolio, please contact the exam office via mail.

Please also note the instructions for using PebblePad & Atlas!

**Good luck creating your portfolio!**